Steps involved in oxidative deamination:

1. In the first step, L-Amino acid oxidases convert the L-amino acid into L-imino acid, which involves FMN as the coenzyme. The resulting peroxide that is formed is detoxified by catalase (Figure 4.7).

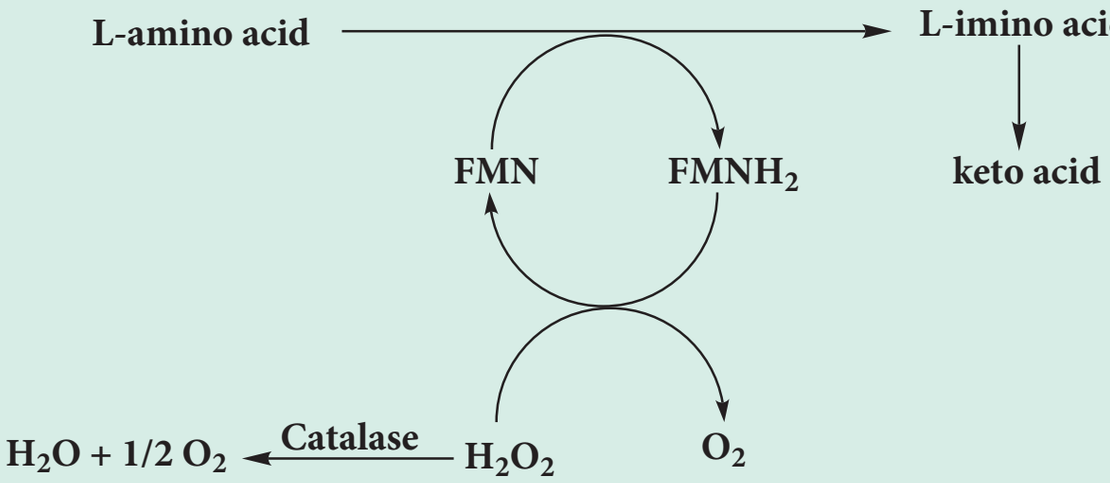

Figure 4.7 Enzymatic reaction catalyzed by L-amino acid oxidase

2. In the second step, the imino acid undergoes hydrolytic cleavage to keto acid and ammonia.

All L-amino acids except hydroxyl amino acids and dicarboxy amino acids can be acted upon by L-amino acid oxidases.

D-amino acid oxidases utilize FAD as the coenzyme.

Non-oxidative deamination

Dehydratases act on hydroxyl amino acids like serine and threonine to form pyruvate and alpha keto butyric acid, respectively.

Cysteine desulfurase removes the sulfydryl group of cysteine to form pyruvate.

The amide groups of glutamine and asparagine are removed by glutaminase and asparaginase, respectively.

4.1.4 Decarboxylation

The removal of carboxyl group from the amino acid in the form of CO_2 is called as decarboxylation. Such decarboxylation reactions are involved in the formation of biologically important amines (Figure 4.8). Several amino acid decarboxylases are present in tissues like liver, kidney, lungs and brain that decarboxylate the respective amino acids to form amines.

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